

Optimization Methods for Support Vector Machines

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1 Introduction

Support Vector Machines (SVMs) are powerful supervised learning models designed for binary classification tasks, where the primary objective is to determine an optimal decision hyperplane that maximizes the geometric margin between classes. Given a dataset of n training samples (x_i, y_i) with features $x_i \in \mathbb{R}^d$ and binary labels $y_i \in \{-1, +1\}$, a soft-margin linear SVM allows for misclassifications in non-linearly separable scenarios by introducing a penalty parameter, $\lambda > 0$. This balances the trade-off between maximizing the margin and minimizing training errors, leading to the standard unconstrained primal optimization problem:

$$\min_{w, b} \frac{1}{2} \|w\|_2^2 + \lambda \sum_{i=1}^n \max\{0, 1 - y_i(x_i^\top w + b)\} \quad (1)$$

where $w \in \mathbb{R}^d$ represents the weight vector, $b \in \mathbb{R}$ is the bias threshold, and the non-smooth hinge loss function $\max\{0, 1 - y_i(x_i^\top w + b)\}$ penalizes margin violations. Optimizing this non-differentiable objective presents unique mathematical challenges, which this report addresses by implementing and evaluating three distinct convex optimization strategies. First, we explore the Alternating Direction Method of Multipliers (ADMM) to solve the primal formulation, decoupling the hinge loss via an auxiliary variable to alternate between solving a closed-form linear system and a proximal soft-thresholding operator. Second, we implement a Block Coordinate Descent (BCD) method to solve the dual formulation, utilizing a baseline Sequential Minimal Optimization (SMO) algorithm that updates dual variables in randomly selected pairs to satisfy the strict linear equality constraint. Finally, to overcome the computational inefficiencies of both first-order baselines, we introduce a Proposed Method based on Newton's method applied to a Huber-smoothed approximation of the primal objective. By replacing the non-differentiable hinge loss with a twice-differentiable surrogate and employing a continuation strategy that progressively sharpens the approximation, this second-order approach achieves quadratic convergence while bypassing the dense Gram matrix required by dual methods. Through rigorous mathematical derivation and empirical testing on synthetic and real-world datasets, this report demonstrates how the proposed method achieves orders-of-magnitude improvement in execution speed and iteration efficiency compared to the baseline approaches.

2 Alternating Direction Method of Multipliers (ADMM)

2.1 Constrained Primal Formulation

The standard unconstrained primal formulation introduces a non-smooth hinge loss term due to the $\max(0, \cdot)$ function, making standard gradient-based optimization methods difficult to apply directly. To isolate this non-smoothness and apply the Alternating Direction Method of Multipliers (ADMM), we decouple the hinge loss from the linear classifier parameters by introducing an auxiliary vector $z \in \mathbb{R}^n$. By establishing the relation $z_i = 1 - y_i(x_i^\top w + b)$ for each training sample, we transform the original unconstrained problem into an equivalent linearly constrained primal optimization problem:

$$\begin{aligned} \min_{w, b, z} \quad & \frac{1}{2} \|w\|_2^2 + \lambda \sum_{i=1}^n \max(0, z_i) \\ \text{s.t.} \quad & z_i = 1 - y_i(x_i^\top w + b), \quad \forall i = 1, \dots, n \end{aligned} \quad (2)$$

This explicit split separates the strictly convex quadratic term governing the margin width ($\frac{1}{2} \|w\|_2^2$) from the non-differentiable penalty components.

2.2 Augmented Lagrangian Derivation

To solve the linearly constrained primal optimization problem, we construct the Augmented Lagrangian. This optimization framework blends the classic method of Lagrange multipliers with a quadratic penalty function that explicitly penalizes any violations of the equality constraints, thereby guaranteeing strict convexity, enforcing numerical stability, and accelerating convergence.

We begin by establishing the standard, unscaled Lagrangian of the constrained system by introducing a vector of Lagrange multipliers (dual variables) $\mu \in \mathbb{R}^n$:

$$\mathcal{L}(w, b, z, \mu) = \frac{1}{2}\|w\|_2^2 + \lambda \sum_{i=1}^n \max(0, z_i) + \sum_{i=1}^n \mu_i (z_i + y_i(x_i^\top w + b) - 1) \quad (3)$$

Next, we add a quadratic penalty term to regularize constraint violations. Introducing a positive penalty tuning parameter $\gamma > 0$, the unscaled Augmented Lagrangian is formulated as:

$$\begin{aligned} \mathcal{L}_\gamma^{\text{unscaled}}(w, b, z, \mu) &= \frac{1}{2}\|w\|_2^2 + \lambda \sum_{i=1}^n \max(0, z_i) + \sum_{i=1}^n \mu_i (z_i + y_i(x_i^\top w + b) - 1) \\ &\quad + \frac{\gamma}{2} \sum_{i=1}^n (z_i + y_i(x_i^\top w + b) - 1)^2 \end{aligned} \quad (4)$$

While mathematically complete, updating variables using the unscaled objective is algebraically cumbersome due to the separate linear dual term and quadratic penalty term. To streamline the block coordinate updates, we derive the more elegant *scaled dual form* by completing the square over the linear and quadratic constraint expressions. For a scalar constraint violation $c_i = z_i + y_i(x_i^\top w + b) - 1$, we isolate and group the terms containing c_i :

$$\mu_i c_i + \frac{\gamma}{2} c_i^2 = \frac{\gamma}{2} \left(c_i^2 + \frac{2}{\gamma} \mu_i c_i \right) \quad (5)$$

To complete the square inside the parenthesis, we add and subtract the squared constant term $\left(\frac{\mu_i}{\gamma}\right)^2$:

$$\begin{aligned} \frac{\gamma}{2} \left(c_i^2 + \frac{2}{\gamma} \mu_i c_i + \left(\frac{\mu_i}{\gamma}\right)^2 - \left(\frac{\mu_i}{\gamma}\right)^2 \right) &= \frac{\gamma}{2} \left(\left(c_i + \frac{\mu_i}{\gamma} \right)^2 - \left(\frac{\mu_i}{\gamma}\right)^2 \right) \\ &= \frac{\gamma}{2} \left(c_i + \frac{\mu_i}{\gamma} \right)^2 - \frac{1}{2\gamma} \mu_i^2 \end{aligned} \quad (6)$$

Substituting the explicit definition of the constraint violation c_i back into this completed square expansion yields the final, structured, scaled dual form of our Augmented Lagrangian:

$$\mathcal{L}_\gamma(w, b, z, \mu) = \frac{1}{2}\|w\|_2^2 + \lambda \sum_{i=1}^n \max(0, z_i) + \frac{\gamma}{2} \sum_{i=1}^n \left(z_i + y_i(x_i^\top w + b) - 1 + \frac{\mu_i}{\gamma} \right)^2 - \frac{1}{2\gamma} \sum_{i=1}^n \mu_i^2 \quad (7)$$

This rigorous derivation explicitly isolates the dual variable vector as a direct shift within the quadratic penalty norm, vastly simplifying the subsequent partial derivatives required for the sequential primal updates.

2.3 ADMM Update Steps

The core mechanic of ADMM relies on optimizing the Augmented Lagrangian sequentially, updating one block of primal variables at a time while keeping the others frozen, followed by a dual variable ascent step. Let k denote the current iteration index.

Step 1: Update w and b

We fix the auxiliary block $z = z^{(k)}$ and the dual multiplier block $\mu = \mu^{(k)}$, and minimize the scaled Augmented Lagrangian $\mathcal{L}_\gamma$ with respect to the hyperplane parameters w and b . To facilitate vector calculus, we express the objective function by isolating the terms containing w and b :

$$\mathcal{L}_\gamma(w, b) = \frac{1}{2}\|w\|_2^2 + \frac{\gamma}{2} \sum_{i=1}^n \left(z_i^{(k)} + y_i x_i^\top w + y_i b - 1 + \frac{\mu_i^{(k)}}{\gamma} \right)^2 + \text{constant} \quad (8)$$

Let $r_i = z_i^{(k)} - 1 + \frac{\mu_i^{(k)}}{\gamma}$ denote a scalar constant collection. The objective function reduces to:

$$\mathcal{L}_\gamma(w, b) = \frac{1}{2}\|w\|_2^2 + \frac{\gamma}{2} \sum_{i=1}^n (y_i x_i^\top w + y_i b + r_i)^2 \quad (9)$$

To eliminate the summation and simplify the gradient derivation, we convert this minimization problem into its equivalent matrix form. We define the feature matrix $X \in \mathbb{R}^{n \times d}$, the label vector $y \in \mathbb{R}^n$, the diagonal label matrix $Y = \text{diag}(y) \in \mathbb{R}^{n \times n}$, and the constant vector $r \in \mathbb{R}^n$. The sum of squares translates perfectly to the squared L_2 -norm of the aggregated vectors:

$$\mathcal{L}_\gamma(w, b) = \frac{1}{2}\|w\|_2^2 + \frac{\gamma}{2}\|YXw + yb + r\|_2^2 \quad (10)$$

Because this function is convex and globally differentiable with respect to w and b , the global minimum is reached where the matrix gradients vanish simultaneously.

First, we take the gradient with respect to the weight vector w using standard matrix calculus:

$$\nabla_w \mathcal{L}_\gamma = w + \gamma(YX)^\top(YXw + yb + r) = 0 \quad (11)$$

Distributing the transposed matrix product $X^\top Y$ yields:

$$w + \gamma X^\top Y^2 Xw + \gamma X^\top Y yb + \gamma X^\top Y r = 0 \quad (12)$$

Because Y is a diagonal matrix of $\{-1, 1\}$, its square is the identity matrix ($Y^2 = I$). Furthermore, by utilizing the property $Yy = \mathbf{1}$, we isolate the variables:

$$(I + \gamma X^\top X) w + (\gamma X^\top \mathbf{1}) b = -\gamma X^\top Y r \quad (13)$$

Second, we take the gradient with respect to the scalar bias b :

$$\nabla_b \mathcal{L}_\gamma = \gamma y^\top(YXw + yb + r) = 0 \quad (14)$$

Distributing the transpose vector $y^\top$ leads to:

$$\gamma y^\top YXw + \gamma y^\top yb + \gamma y^\top r = 0 \quad (15)$$

Using the inner product identity $y^\top y = n$ and the property $y^\top Y = \mathbf{1}^\top$ to maintain structural symmetry, we obtain the second condition:

$$(\gamma \mathbf{1}^\top X) w + (\gamma n) b = -\gamma y^\top r \quad (16)$$

Finally, by parsing these two isolated matrix equations into a single coupled system of linear blocks, we construct the exact stationary condition:

$$\begin{bmatrix} I + \gamma X^\top X & \gamma X^\top \mathbf{1} \\ \gamma \mathbf{1}^\top X & \gamma n \end{bmatrix} \begin{bmatrix} w^{(k+1)} \\ b^{(k+1)} \end{bmatrix} = \begin{bmatrix} -\gamma X^\top Y r \\ -\gamma y^\top r \end{bmatrix} \quad (17)$$

By solving this symmetric linear block matrix via optimized matrix factorization techniques, we evaluate the exact global primal updates $w^{(k+1)}$ and $b^{(k+1)}$ simultaneously.

Step 2: Update z

Next, we freeze the newly computed parameters $w = w^{(k+1)}$ and $b = b^{(k+1)}$, alongside the dual variables $\mu = \mu^{(k)}$, and minimize the Augmented Lagrangian $\mathcal{L}_\gamma$ with respect to the auxiliary vector z . Isolating the terms that depend on z and pulling out the common summation yields:

$$\mathcal{L}_\gamma(z) = \sum_{i=1}^n \left[\lambda \max(0, z_i) + \mu_i^{(k)} \left(z_i + y_i(x_i^\top w^{(k+1)} + b^{(k+1)}) - 1 \right) + \frac{\gamma}{2} \left(z_i + y_i(x_i^\top w^{(k+1)} + b^{(k+1)}) - 1 \right)^2 \right]$$

By factoring out $\frac{\gamma}{2}$ and completing the square inside the summation, we can absorb the linear dual multiplier term into the quadratic penalty. Dropping the remaining constant terms that do not affect the argmin, the high-dimensional objective elegantly decouples into n independent, univariate coordinate minimizations:

$$\min_{z_i} \lambda \max(0, z_i) + \frac{\gamma}{2} \left(z_i - \left[1 - y_i(x_i^\top w^{(k+1)} + b^{(k+1)}) - \frac{\mu_i^{(k)}}{\gamma} \right] \right)^2$$

We define the intermediate scalar constant collection as $v_i = 1 - y_i(x_i^\top w^{(k+1)} + b^{(k+1)}) - \frac{\mu_i^{(k)}}{\gamma}$. Dividing by the positive scalar γ , the problem reduces to evaluating the standard proximal operator of the scaled hinge loss:

$$\min_{z_i} f(z_i) = \frac{\lambda}{\gamma} \max(0, z_i) + \frac{1}{2} (z_i - v_i)^2$$

Because the $\max(0, z_i)$ function is non-differentiable at $z_i = 0$, we find its global minimum by identifying where the subgradient vanishes ($0 \in \partial f(z_i)$). The subgradient set of the max function is 1 for $z_i > 0$, 0 for $z_i < 0$, and the continuous interval $[0, 1]$ at $z_i = 0$. Analyzing the optimality condition across these three distinct domains yields:

Case 1: $z_i < 0$

The max term evaluates to 0, leaving a strictly differentiable function:

$$\frac{\partial f}{\partial z_i} = z_i - v_i = 0 \implies z_i = v_i$$

For the structural assumption $z_i < 0$ to hold true, this domain requires $v_i < 0$.

Case 2: $z_i > 0$

The max term evaluates to z_i , making its derivative equal to 1:

$$\frac{\partial f}{\partial z_i} = \frac{\lambda}{\gamma} + z_i - v_i = 0 \implies z_i = v_i - \frac{\lambda}{\gamma}$$

For the structural assumption $z_i > 0$ to hold true, this domain requires $v_i - \frac{\lambda}{\gamma} > 0 \implies v_i > \frac{\lambda}{\gamma}$.

Case 3: $z_i = 0$

At the non-differentiable point, the vanishing subgradient condition requires that zero belongs to the resulting set:

$$0 \in \frac{\lambda}{\gamma} [0, 1] + (0 - v_i) \implies v_i \in \left[0, \frac{\lambda}{\gamma} \right]$$

Consequently, if v_i falls within this closed interval, the optimal coordinate solution is pinned exactly at $z_i = 0$.

Synthesizing these three mutually exclusive cases yields the exact closed-form update via the soft-thresholding function:

$$z_i^{(k+1)} = \begin{cases} v_i & \text{if } v_i < 0 \\ v_i - \frac{\lambda}{\gamma} & \text{if } v_i > \frac{\lambda}{\gamma} \\ 0 & \text{if } 0 \leq v_i \leq \frac{\lambda}{\gamma} \end{cases}$$

Step 3: Update μ

Finally, the dual multiplier block μ is updated by performing a gradient ascent step on the dual problem to maximize the Augmented Lagrangian. Following standard multiplier method theory, the step size for the dual update is set precisely equal to the penalty parameter γ . Isolating the terms related to the dual step, the update for a single coordinate index i corresponds directly to its constraint residual:

$$\mu_i^{(k+1)} = \mu_i^{(k)} + \gamma \left(z_i^{(k+1)} + y_i \left(x_i^\top w^{(k+1)} + b^{(k+1)} \right) - 1 \right) \quad (18)$$

By stacking these independent coordinate evaluations across all n samples, we convert the iteration rule into an efficient, fully vectorized form. We utilize our established dataset definitions, including the feature matrix $X \in \mathbb{R}^{n \times d}$, the label vector $y \in \mathbb{R}^n$, the diagonal label matrix $Y = \text{diag}(y) \in \mathbb{R}^{n \times n}$, and the column vector of ones $\mathbf{1} \in \mathbb{R}^n$.

Because $y_i x_i^\top w$ maps to the matrix-vector product YXw , and the scalar bias distribution $y_i b$ maps to $y b$, the complete system of dual parameters updates simultaneously via:

$$\mu^{(k+1)} = \mu^{(k)} + \gamma \left(z^{(k+1)} + YXw^{(k+1)} + yb^{(k+1)} - \mathbf{1} \right) \quad (19)$$

This step effectively penalizes any remaining mismatch between the structural margins of the hyperplane and the non-negative margins captured by the auxiliary slack vector $z^{(k+1)}$, driving the system toward primal-dual feasibility.

2.4 Computational Cost Analysis

The ADMM framework achieves efficiency through upfront pre-computation. Prior to the main loop, the algorithm constructs the static block coefficient matrix:

$$A = \begin{bmatrix} I + \gamma X^\top X & \gamma X^\top \mathbf{1} \\ \gamma \mathbf{1}^\top X & \gamma n \end{bmatrix} \in \mathbb{R}^{(d+1) \times (d+1)} \quad (20)$$

Computing $X^\top X$ costs $\mathcal{O}(nd^2)$. Because A is symmetric positive-definite, we compute its Cholesky factorization ($A = LL^\top$) once at a cost of $\mathcal{O}(d^3)$, removing matrix inversions from the iterative loop. The per-iteration costs are:

- **Step 1 (Primal w, b Update):** Computing the RHS vector costs $\mathcal{O}(nd)$. Solving the pre-factored system via forward/backward substitution costs $\mathcal{O}(d^2)$.
- **Step 2 (Auxiliary z Update):** Computing $v = \mathbf{1} - Y(Xw + b) - \mu/\gamma$ costs $\mathcal{O}(nd)$ (dominated by Xw). The proximal soft-thresholding is element-wise in $\mathcal{O}(n)$.
- **Step 3 (Dual μ Update):** Reuses the cached margin products from Step 2, requiring only $\mathcal{O}(n)$ element-wise arithmetic.

When integrated, the total computational cost per iteration simplifies to $\mathcal{O}(nd + d^2)$. Since $d \ll n$ for both datasets ($d \leq 50$, $n \geq 500$), the per-iteration cost is dominated by $\mathcal{O}(nd)$.

Algorithm 1 ADMM for Primal SVM

Require: $X \in \mathbb{R}^{n \times d}$, $y \in \{-1, +1\}^n$, $\lambda > 0$, $\gamma > 0$, tolerance ϵ , max iterations K

Ensure: $w^* \in \mathbb{R}^d$, $b^* \in \mathbb{R}$

```
1: Initialize  $w \leftarrow 0$ ,  $b \leftarrow 0$ ,  $z \leftarrow 0$ ,  $\mu \leftarrow 0$ 
2: Precompute coefficient matrix  $A \in \mathbb{R}^{(d+1) \times (d+1)}$   $\triangleright \mathcal{O}(nd^2)$  one-time cost
3: for  $k = 1, 2, \dots, K$  do
4:    $r \leftarrow z - \mathbf{1} + \mu/\gamma$ 
5:   Solve  $A[w; b]^\top = [-\gamma X^\top Y r; -\gamma y^\top r]$  for  $(w, b)$ 
6:    $v \leftarrow \mathbf{1} - (YXw + yb) - \mu/\gamma$ 
7:   Update  $z$  element-wise:  $z_i = v_i$  if  $v_i < 0$ ;  $z_i = 0$  if  $0 \leq v_i \leq \frac{\lambda}{\gamma}$ ;  $z_i = v_i - \frac{\lambda}{\gamma}$  otherwise
8:    $\mu \leftarrow \mu + \gamma(z + YXw + yb - \mathbf{1})$ 
9:   if  $\|r_{\text{primal}}\| < \epsilon$  and  $\|r_{\text{dual}}\| < \epsilon$  then
10:    break
11:  end if
12: end for
13: return  $w, b$ 
```

3 Block Coordinate Descent (BCD) for the Dual SVM

3.1 Dual Formulation Derivation and Optimality Conditions

To derive the dual formulation of the soft-margin linear Support Vector Machine, we begin directly from the linearly constrained primal optimization problem established for our ADMM framework:

$$\begin{aligned} \min_{w, b, z} \quad & \frac{1}{2} \|w\|_2^2 + \lambda \sum_{i=1}^n \max(0, z_i) \\ \text{s.t.} \quad & z_i = 1 - y_i(x_i^\top w + b), \quad \forall i = 1, \dots, n \end{aligned} \quad (21)$$

To evaluate this dual transition rigorously, we must first verify that strong duality holds. We appeal to Slater's Condition for convex optimization, which dictates that if the primal objective function is convex and all domain constraints are affine, strong duality is satisfied if there exists at least one feasible point. Because our equality constraints $z_i + y_i(x_i^\top w + b) - 1 = 0$ are entirely linear with respect to the primal blocks, Slater's condition holds. This guarantees a duality gap of exactly zero, ensuring that the global maximum of the dual problem matches the global minimum of the primal problem precisely (min Primal = max Dual).

With strong duality guaranteed, the global optimizer can be completely characterized by the Karush-Kuhn-Tucker (KKT) conditions. We construct the primal Lagrangian function $\mathcal{L}(w, b, z, \alpha)$ by introducing a vector of dual Lagrange multipliers $\alpha \in \mathbb{R}^n$ to couple the linear equality constraints back into the objective:

$$\mathcal{L}(w, b, z, \alpha) = \frac{1}{2} \|w\|_2^2 + \lambda \sum_{i=1}^n \max(0, z_i) + \sum_{i=1}^n \alpha_i (1 - y_i(x_i^\top w + b) - z_i) \quad (22)$$

Distributing the summation terms allows us to segregate the independent primal blocks cleanly:

$$\mathcal{L}(w, b, z, \alpha) = \frac{1}{2} \|w\|_2^2 - \sum_{i=1}^n \alpha_i y_i x_i^\top w - \left(\sum_{i=1}^n \alpha_i y_i \right) b + \sum_{i=1}^n \alpha_i + \sum_{i=1}^n (\lambda \max(0, z_i) - \alpha_i z_i) \quad (23)$$

By Lagrange duality, the first-order necessary and sufficient KKT stationarity conditions require the partial gradients of the Lagrangian with respect to the continuous primal variables to vanish at the saddle point.

First, evaluating the stationarity condition with respect to the weight vector w yields:

$$\nabla_w \mathcal{L} = w - \sum_{i=1}^n \alpha_i y_i x_i = 0 \implies w^* = \sum_{i=1}^n \alpha_i y_i x_i \quad (24)$$

This serves as the critical primal-dual bridge relationship, proving that the optimal hyperplane orientation is formed by a linear combination of the support vector features.

Second, evaluating the stationarity condition with respect to the scalar bias b yields:

$$\frac{\partial \mathcal{L}}{\partial b} = -\sum_{i=1}^n \alpha_i y_i = 0 \implies \sum_{i=1}^n \alpha_i y_i = 0 \quad (25)$$

This establishes a strict zero-sum constraint on the scaled dual coordinates, preventing the dual objective function from spiraling to negative infinity.

Third, we examine the stationarity condition with respect to the auxiliary slack variables z_i . Because the term $\max(0, z_i)$ is non-differentiable at the origin, we invoke subgradient calculus. The KKT stationarity condition requires that zero belongs to the resulting subgradient set:

$$0 \in \partial_{z_i} \mathcal{L} \implies 0 \in \lambda \partial \max(0, z_i) - \alpha_i \quad (26)$$

Recalling that the subgradient of $\max(0, z_i)$ maps to 0 for $z_i < 0$, 1 for $z_i > 0$, and the bounded interval $[0, 1]$ at $z_i = 0$, the condition dictates that α_i must reside inside the scaled subgradient bounds. This elegantly derives the classic dual box constraints:

$$\alpha_i \in \lambda[0, 1] \implies 0 \leq \alpha_i \leq \lambda, \quad \forall i = 1, \dots, n \quad (27)$$

Under these active box constraints, the optimization over z_i resolves, and the remaining slack summation evaluates exactly to zero ($\sum_{i=1}^n [\lambda \max(0, z_i) - \alpha_i z_i] = 0$).

To eliminate the primal parameters entirely, we substitute our optimal updates and active constraints back into the expanded Lagrangian:

$$\begin{aligned} W(\alpha) &= \frac{1}{2} \left\| \sum_{i=1}^n \alpha_i y_i x_i \right\|_2^2 - \left(\sum_{i=1}^n \alpha_i y_i x_i \right)^\top \left(\sum_{j=1}^n \alpha_j y_j x_j \right) - (0)b + \sum_{i=1}^n \alpha_i + 0 \\ &= \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^\top x_j - \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^\top x_j + \sum_{i=1}^n \alpha_i \\ &= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^\top x_j \end{aligned} \quad (28)$$

Maximizing this remaining function subject to our derived bounds yields the final, standard Dual Support Vector Machine optimization problem:

$$\begin{aligned} \max_{\alpha} \quad & W(\alpha) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^\top x_j \\ \text{s.t.} \quad & \sum_{i=1}^n \alpha_i y_i = 0 \\ & 0 \leq \alpha_i \leq \lambda, \quad \forall i = 1, \dots, n \end{aligned} \quad (29)$$

This dual problem forms the exact foundation for our Block Coordinate Descent (BCD) Algorithm.

3.2 SMO Update Steps Derivation

The Sequential Minimal Optimization (SMO) algorithm optimizes a selected pair of dual multipliers, α_i and α_j , while holding the remaining $n - 2$ variables constant. To streamline the mathematical derivation, we define the inner product between two feature vectors as $K_{ij} = x_i^\top x_j$.

We isolate the components of the dual objective function $W(\alpha)$ that explicitly depend on our active pair:

$$\begin{aligned} W(\alpha_i, \alpha_j) = & \alpha_i + \alpha_j - \frac{1}{2}\alpha_i^2 K_{ii} - \frac{1}{2}\alpha_j^2 K_{jj} - \alpha_i \alpha_j y_i y_j K_{ij} \\ & - \alpha_i y_i \sum_{m \neq i, j}^n \alpha_m y_m K_{mi} - \alpha_j y_j \sum_{m \neq i, j}^n \alpha_m y_m K_{mj} + \text{constant} \end{aligned} \quad (30)$$

Let $f(x) = \sum_{m=1}^n \alpha_m y_m K_{mx} + b$ represent the model's forward prediction function. We express the isolated remaining summation terms using the raw classification errors $E_i = f(x_i) - y_i$ and $E_j = f(x_j) - y_j$ evaluated at the current iteration k :

$$\sum_{m \neq i, j}^n \alpha_m y_m K_{mi} = E_i + y_i - b - \alpha_i^{(k)} y_i K_{ii} - \alpha_j^{(k)} y_j K_{ji} \quad (31)$$

$$\sum_{m \neq i, j}^n \alpha_m y_m K_{mj} = E_j + y_j - b - \alpha_i^{(k)} y_i K_{ij} - \alpha_j^{(k)} y_j K_{jj} \quad (32)$$

Because the active pair must strictly satisfy the linear equality constraint at all times, the updates are coupled by the relationship $\alpha_i y_i + \alpha_j y_j = -\sum_{m \neq i, j}^n \alpha_m y_m = \zeta$, where ζ is a scalar constant. Multiplying by y_i and noting $y_i^2 = 1$, we express α_i entirely as a linear function of α_j :

$$\alpha_i = y_i (\zeta - \alpha_j y_j) \implies \alpha_i = \alpha_i^{(k)} + y_i y_j (\alpha_j^{(k)} - \alpha_j) \quad (33)$$

Substituting this linear dependency back into $W(\alpha_i, \alpha_j)$ transforms the joint optimization into a univariate optimization problem with respect to α_j . We define the objective function's directional curvature parameter as $\eta = 2K_{ij} - K_{ii} - K_{jj}$. Differentiating the substituted objective and setting $\frac{\partial W}{\partial \alpha_j} = 0$ yields the clean analytical unclipped maximum:

$$\alpha_j^{\text{unclipped}} = \alpha_j^{(k)} - \frac{y_j(E_i - E_j)}{\eta} \quad (34)$$

To enforce the dual box constraints $0 \leq \alpha \leq \lambda$, this unclipped value must be projected onto the valid segment of the constraint line. We define the lower bound L and upper bound H based on label parity:

- **Case 1:** $y_i \neq y_j$. The constraint line takes the form $\alpha_j - \alpha_i = \zeta$. The bounding limits are:

$$L = \max(0, \alpha_j^{(k)} - \alpha_i^{(k)}), \quad H = \min(\lambda, \lambda + \alpha_j^{(k)} - \alpha_i^{(k)}) \quad (35)$$

- **Case 2:** $y_i = y_j$. The constraint line takes the form $\alpha_j + \alpha_i = \zeta$. The bounding limits are:

$$L = \max(0, \alpha_i^{(k)} + \alpha_j^{(k)} - \lambda), \quad H = \min(\lambda, \alpha_i^{(k)} + \alpha_j^{(k)}) \quad (36)$$

Clipping the analytical unclipped update inside these boundaries provides the valid updated coordinate for the second variable:

$$\alpha_j^{(k+1)} = \begin{cases} H & \text{if } \alpha_j^{\text{unclipped}} \geq H \\ \alpha_j^{\text{unclipped}} & \text{if } L < \alpha_j^{\text{unclipped}} < H \\ L & \text{if } \alpha_j^{\text{unclipped}} \leq L \end{cases} \quad (37)$$

Once $\alpha_j^{(k+1)}$ is locked, the first variable is updated linearly to balance the sum-to-zero constraint:

$$\alpha_i^{(k+1)} = \alpha_i^{(k)} + y_i y_j (\alpha_j^{(k)} - \alpha_j^{(k+1)}) \quad (38)$$

Finally, to fulfill the Karush-Kuhn-Tucker complementary slackness conditions, the bias threshold b must be repositioned. If $0 < \alpha_i^{(k+1)} < \lambda$, the sample lies directly on the margin, forcing $b^{(k+1)} = b_1$. If $0 < \alpha_j^{(k+1)} < \lambda$, the sample lies directly on the margin, forcing $b^{(k+1)} = b_2$. These prospective shifts are derived via the active support coordinates as:

$$b_1 = b^{(k)} - E_i - y_i \left(\alpha_i^{(k+1)} - \alpha_i^{(k)} \right) K_{ii} - y_j \left(\alpha_j^{(k+1)} - \alpha_j^{(k)} \right) K_{ij} \quad (39)$$

$$b_2 = b^{(k)} - E_j - y_i \left(\alpha_i^{(k+1)} - \alpha_i^{(k)} \right) K_{ij} - y_j \left(\alpha_j^{(k+1)} - \alpha_j^{(k)} \right) K_{jj} \quad (40)$$

If both updates sit on the box boundaries (0 or λ), the algorithm sets the new threshold to their mid-point: $b^{(k+1)} = \frac{b_1 + b_2}{2}$.

3.3 Computational Cost Analysis

The BCD algorithm precomputes the Gram matrix $K = XX^\top \in \mathbb{R}^{n \times n}$ at a one-time cost of $\mathcal{O}(n^2d)$ and $\mathcal{O}(n^2)$ memory. Once cached, computing the prediction errors E_i, E_j for a selected pair requires $\mathcal{O}(n)$ time (a single row lookup and summation). The clipping, coordinate updates, and bias adjustments are $\mathcal{O}(1)$ scalar operations. Thus, a single pair update costs $\mathcal{O}(n)$, and a full epoch over all n samples costs $\mathcal{O}(n^2)$.

While this avoids cubic dependencies on d , the $\mathcal{O}(n^2)$ memory for the Gram matrix becomes a bottleneck for large sample sizes ($n \gg 10^5$).

Algorithm 2 Block Coordinate Descent (SMO) for Dual SVM

Require: $X \in \mathbb{R}^{n \times d}$, $y \in \{-1, +1\}^n$, $\lambda > 0$, max epochs T

Ensure: $\alpha^* \in \mathbb{R}^n$, $b^* \in \mathbb{R}$

```

1: Initialize  $\alpha \leftarrow 0$ ,  $b \leftarrow 0$ 
2: Precompute Gram matrix  $K = XX^\top \in \mathbb{R}^{n \times n}$   $\triangleright \mathcal{O}(n^2d)$ 
3: for epoch = 1, 2, ...,  $T$  do
4:   changed  $\leftarrow 0$ 
5:   for  $i = 1, \dots, n$  do
6:     Select  $j \neq i$  randomly
7:     Compute errors  $E_i = f(x_i) - y_i$ ,  $E_j = f(x_j) - y_j$ 
8:     Compute curvature  $\eta = 2K_{ij} - K_{ii} - K_{jj}$ 
9:     if  $\eta \geq 0$  or  $L = H$  then
10:      continue
11:    end if
12:     $\alpha_j^{\text{new}} \leftarrow \text{clip}(\alpha_j - y_j(E_i - E_j)/\eta, L, H)$ 
13:     $\alpha_i^{\text{new}} \leftarrow \alpha_i + y_i y_j (\alpha_j^{\text{old}} - \alpha_j^{\text{new}})$ 
14:    Update bias  $b$  using KKT conditions
15:    changed  $\leftarrow$  changed + 1
16:  end for
17:  if changed = 0 then
18:    break
19:  end if
20: end for
21: return  $\alpha$ ,  $b$ 
```

4 Proposed Method for Performance Enhancement

4.1 Limitations of First-Order and Dual Solvers

While the baseline Block Coordinate Descent (BCD) and ADMM algorithms successfully optimize the Support Vector Machine (SVM) objective, they are inherently limited by their first-order nature. ADMM relies

on subgradient steps, while dual coordinate methods take localized 1D steps. Both approaches yield linear convergence rates characterized by zigzagging trajectories across the objective landscape. Furthermore, dual methods require caching a dense $\mathcal{O}(n^2)$ Gram matrix, severely bottlenecking scalability for large datasets.

To break through these computational limits, we propose shifting the optimization back to the primal domain and utilizing second-order curvature information via Newton’s method. A true second-order method achieves quadratic convergence, drastically reducing the required number of iterations. However, standard Newton’s method cannot be applied directly to the primal SVM formulation because the standard hinge loss function, $\max(0, 1 - t)$, contains a sharp, non-differentiable corner at $t = 1$, rendering the Hessian matrix undefined.

4.2 Huber-Smoothed Primal Formulation

To establish a twice-differentiable objective function, we propose replacing the exact hinge loss with a $\mathcal{C}^1$ -smooth approximation using a parameterized Huber function [2, 1]. Let the margin residual for a given sample be defined as $r_i = 1 - y_i(x_i^\top w + b)$. We define the Huber-smoothed hinge loss $\phi_\mu(r_i)$, controlled by a smoothing parameter $\mu > 0$, as:

$$\phi_\mu(r_i) = \begin{cases} 0 & \text{if } r_i \leq 0 \\ \frac{r_i^2}{2\mu} & \text{if } 0 < r_i \leq \mu \\ r_i - \frac{\mu}{2} & \text{if } r_i > \mu \end{cases} \quad (41)$$

This piecewise function seamlessly connects the flat zero-loss region ($r_i \leq 0$) to the linear active-loss region ($r_i > \mu$) using a smooth quadratic transition zone ($0 < r_i \leq \mu$). The modified, fully differentiable primal objective function becomes:

$$\mathcal{J}_\mu(w, b) = \frac{1}{2} \|w\|_2^2 + \lambda \sum_{i=1}^n \phi_\mu(r_i) \quad (42)$$

4.3 Second-Order Optimization via Newton’s Method

Because $\mathcal{J}_\mu(w, b)$ is twice-differentiable, we can analytically compute its exact gradient and Hessian to execute Newton steps. We first derive the first derivative of the smoothed loss function, $\phi'_\mu(r_i)$:

$$\phi'_\mu(r_i) = \begin{cases} 0 & \text{if } r_i \leq 0 \\ \frac{r_i}{\mu} & \text{if } 0 < r_i \leq \mu \\ 1 & \text{if } r_i > \mu \end{cases} \quad (43)$$

Applying the chain rule, the gradients with respect to the weights w and bias b are formulated as:

$$\nabla_w \mathcal{J}_\mu = w - \lambda \sum_{i=1}^n y_i x_i \phi'_\mu(r_i) \quad (44)$$

$$\nabla_b \mathcal{J}_\mu = -\lambda \sum_{i=1}^n y_i \phi'_\mu(r_i) \quad (45)$$

Next, we extract the curvature by evaluating the second derivative $\phi''_\mu(r_i)$. The second derivative is strictly zero everywhere except inside the quadratic transition zone:

$$\phi''_\mu(r_i) = \begin{cases} \frac{1}{\mu} & \text{if } 0 < r_i \leq \mu \\ 0 & \text{otherwise} \end{cases} \quad (46)$$

To construct the full Hessian matrix $H \in \mathbb{R}^{(d+1) \times (d+1)}$, we compute the second-order partial derivatives of the objective function. Applying the chain rule to the gradients yields the sub-matrix blocks H_{ww} , H_{wb} , and

H_{bb} . Since the class labels are $y_i \in \{-1, 1\}$, the squared labels naturally simplify to $y_i^2 = 1$:

$$H_{ww} = \frac{\partial \nabla_w \mathcal{J}_\mu}{\partial w} = I + \lambda \sum_{i=1}^n \phi''_\mu(r_i) (-y_i x_i) (-y_i x_i^\top) = I + \lambda \sum_{i=1}^n \phi''_\mu(r_i) x_i x_i^\top \quad (47)$$

$$H_{wb} = \frac{\partial \nabla_w \mathcal{J}_\mu}{\partial b} = 0 + \lambda \sum_{i=1}^n \phi''_\mu(r_i) (-y_i x_i) (-y_i) = \lambda \sum_{i=1}^n \phi''_\mu(r_i) x_i \quad (48)$$

$$H_{bb} = \frac{\partial \nabla_b \mathcal{J}_\mu}{\partial b} = 0 + \lambda \sum_{i=1}^n \phi''_\mu(r_i) (-y_i) (-y_i) = \lambda \sum_{i=1}^n \phi''_\mu(r_i) \quad (49)$$

Notice that $\phi''_\mu(r_i)$ acts as a sparse gating mechanism: the curvature is exactly $\frac{1}{\mu}$ when $0 < r_i \leq \mu$, and 0 everywhere else. Let $\mathcal{S}$ denote the dynamic active set of indices where the residuals fall strictly within this transition zone, and let $n_{\mathcal{S}}$ be the number of active samples. By filtering out the zero-curvature samples, we avoid inefficient loops and vectorize the summations using the active feature sub-matrix $X_{\mathcal{S}} \in \mathbb{R}^{n_{\mathcal{S}} \times d}$ and a vector of ones $\mathbf{1} \in \mathbb{R}^{n_{\mathcal{S}} \times 1}$. This substitution yields the highly optimized block Hessian:

$$H = \begin{bmatrix} H_{ww} & H_{wb} \\ H_{wb}^\top & H_{bb} \end{bmatrix} = \begin{bmatrix} I + \frac{\lambda}{\mu} X_{\mathcal{S}}^\top X_{\mathcal{S}} & \frac{\lambda}{\mu} X_{\mathcal{S}}^\top \mathbf{1} \\ \frac{\lambda}{\mu} \mathbf{1}^\top X_{\mathcal{S}} & \frac{\lambda}{\mu} n_{\mathcal{S}} \end{bmatrix} \quad (50)$$

With the exact gradient vector $g = [\nabla_w \mathcal{J}_\mu; \nabla_b \mathcal{J}_\mu]$ and the fully assembled Hessian H , the algorithm calculates the optimal Newton descent direction $\Delta = -H^{-1}g$. To ensure global convergence and prevent overshooting the quadratic basin, a standard Armijo backtracking line search is applied along this trajectory to determine the optimal step size α . The parameters are then updated simultaneously:

$$\begin{bmatrix} w^{(k+1)} \\ b^{(k+1)} \end{bmatrix} = \begin{bmatrix} w^{(k)} \\ b^{(k)} \end{bmatrix} + \alpha \Delta \quad (51)$$

4.4 Theoretical Performance Advantages

The proposed method improves upon the baselines through three key advantages:

- **Quadratic Convergence Rate:** By incorporating exact Hessian information, Newton’s method achieves quadratic convergence near the optimum, reducing required iterations by orders of magnitude compared to the linear convergence of ADMM and BCD.
- **Elimination of the Gram Matrix:** BCD requires an $n \times n$ kernel matrix ($\mathcal{O}(n^2)$ memory, $\mathcal{O}(n^2 d)$ setup). Operating in the primal space, the proposed method requires only $\mathcal{O}(nd)$ storage.
- **Robustness via Continuation:** Warm-starting across a decaying μ schedule avoids the tradeoff between optimization speed (large μ) and solution accuracy (small μ), ensuring rapid convergence to the exact SVM solution.

4.5 Computational Complexity and Cost Analysis

Per Newton iteration, the dominant costs are:

- **Residuals and gradient:** Computing $r = \mathbf{1} - y \odot (Xw + b)$ and the gradient g each cost $\mathcal{O}(nd)$.
- **Hessian construction:** Building $X_{\mathcal{S}}^\top X_{\mathcal{S}}$ over the $n_{\mathcal{S}} \leq n$ active samples costs $\mathcal{O}(n_{\mathcal{S}} d^2)$.
- **Newton solve:** The Hessian is $(d+1) \times (d+1)$, so solving $\Delta = -H^{-1}g$ costs $\mathcal{O}(d^3)$.
- **Line search:** A constant number of objective evaluations at $\mathcal{O}(nd)$ each.

The overall per-iteration complexity is $\mathcal{O}(n_{\mathcal{S}} d^2 + d^3)$ time and $\mathcal{O}(nd + d^2)$ memory. In the worst case ($n_{\mathcal{S}} = n$), this becomes $\mathcal{O}(nd^2)$ per iteration. However, as the algorithm converges and μ decreases, the active set $\mathcal{S}$ shrinks considerably, reducing the effective per-iteration cost. Combined with the quadratic convergence rate (far fewer iterations), this explains the sub-second runtimes observed empirically, compared to the $\mathcal{O}(n^2)$ per-epoch cost of BCD.

4.6 Continuation Strategy and Warm-Starting

A large μ yields a well-conditioned Hessian but optimizes a loose approximation; a small μ closely approximates the true hinge loss but can cause ill-conditioning. The continuation strategy resolves this tradeoff by starting with $\mu = 1.0$ and decaying through a logarithmic schedule $\mu \in \{1.0, 0.1, 0.01, 0.001\}$. At each stage, the converged parameters from the previous stage serve as the warm-start initialization, ensuring the iterates remain within the quadratic basin of convergence as the landscape sharpens.

Convergence to the Exact Problem: A key theoretical property of the Huber-smoothed hinge loss is that it converges uniformly to the original non-smooth hinge loss as $\mu \rightarrow 0$. Specifically, for any margin residual r_i , the pointwise approximation error satisfies:

$$0 \leq \phi_\mu(r_i) - \max(0, r_i) \leq \frac{\mu}{2}, \quad \forall r_i \in \mathbb{R} \quad (52)$$

This bound guarantees that $\mathcal{J}_\mu(w, b) \rightarrow \frac{1}{2}\|w\|_2^2 + \lambda \sum_{i=1}^n \max(0, r_i)$ uniformly as $\mu \rightarrow 0$. Consequently, the sequence of minimizers $\{(w_\mu^*, b_\mu^*)\}$ converges to the true SVM solution as the smoothing vanishes. In our implementation, the final stage uses $\mu = 0.001$, bounding the maximum per-sample approximation error to 5×10^{-4} , which is well below the convergence tolerances of the baseline methods.

Objective Evaluation Note: While the algorithm solves the smooth surrogate $\mathcal{J}_\mu$ to efficiently find the optimal parameters, all final objective values and training metrics reported in the subsequent experiments are computed strictly using the original, non-smooth exact SVM objective. This ensures a fair and fully equivalent comparison against all baseline solvers.

Algorithm 3 Huber-Smoothed Newton with Continuation

Require: $X \in \mathbb{R}^{n \times d}$, $y \in \{-1, +1\}^n$, $\lambda > 0$, schedule $(\mu_1, \dots, \mu_M)$, tolerance ϵ , max iterations K

Ensure: $w^* \in \mathbb{R}^d$, $b^* \in \mathbb{R}$

```

1: Initialize  $w \leftarrow 0$ ,  $b \leftarrow 0$ 
2: for each  $\mu$  in  $(\mu_1, \dots, \mu_M)$  do ▷ Continuation loop
3:   for  $k = 1, 2, \dots, K$  do
4:     Compute residuals  $r_i = 1 - y_i(x_i^\top w + b)$  for all  $i$ 
5:     Identify active set  $\mathcal{S} = \{i : 0 < r_i \leq \mu\}$ 
6:     Compute gradient  $g \in \mathbb{R}^{d+1}$  using  $\phi'_\mu(r)$ 
7:     if  $\|g\| < \epsilon$  then
8:       break ▷ Converged for current  $\mu$ 
9:     end if
10:    Build Hessian  $H \in \mathbb{R}^{(d+1) \times (d+1)}$  using active set  $X_{\mathcal{S}}$ 
11:    Solve Newton direction  $\Delta = -H^{-1}g$ 
12:    Armijo backtracking line search: find step size  $\alpha$ 
13:     $[w; b] \leftarrow [w; b] + \alpha\Delta$ 
14:  end for
15: end for
16: return  $w, b$ 
```

5 Experimental Setup

5.1 Datasets

To rigorously evaluate and compare the numerical performance of the optimization algorithms, empirical testing is conducted across two distinct classification benchmarks:

- **Synthetic Dataset:** A controlled dataset of $n = 1000$ training samples and $d = 50$ features is generated using the provided data-generation code. The classes are distributed with a condition number of 100 to control the optimization difficulty. Following reproducibility requirements, no standardization or normalization is applied to the synthetic features. Labels are mapped to $y_i \in \{-1, +1\}$.

- **Breast Cancer Dataset:** A real-world medical benchmark consisting of $n = 569$ samples and $d = 30$ features. The features represent geometric characteristics of cell nuclei. The dataset is preprocessed by mapping the target labels to $y_i \in \{-1, +1\}$ (+1 for malignant, -1 for benign). Features are standardized using only the training-set mean and standard deviation, with the identical transformation applied to the test set.

For both environments, a stratified 70/30 train-test split is implemented to ensure that the class proportions are maintained across the training and validation subsets.

5.2 Evaluation Metrics

The algorithms are tracked and cross-examined at the completion of every iteration (or epoch) using three standard mathematical metrics calculated strictly outside of the timed optimization loops:

1. **Primal Objective Value:** Evaluated directly via the primal loss function:

$$P(w, b) = \frac{1}{2} \|w\|_2^2 + \lambda \sum_{i=1}^n \max\{0, 1 - y_i(x_i^\top w + b)\} \quad (53)$$

For the dual solver (BCD), the primal parameters are reconstructed dynamically at each evaluation step using the primal-dual bridge $w = \sum_{i=1}^n \alpha_i y_i x_i$.

2. **Test Accuracy:** The generalization power of the current hyperplane configuration is verified on the unseen validation partition:

$$\text{Accuracy} = \frac{1}{n_{test}} \sum_{m=1}^{n_{test}} \mathbb{I}(\text{sign}(x_m^\top w + b) == y_m) \quad (54)$$

where $\mathbb{I}(\cdot)$ represents the indicator function.

3. **Cumulative Optimization Time:** Measured in CPU seconds using precise monotonic timestamps tracking only the active optimization loops. Initializations, data loading, and metric evaluations are omitted from this timer to ensure fair competition.

5.3 Implementation Details

All experiments use a fixed penalty parameter $\lambda = 1$. The stopping tolerances and maximum iteration limits for each algorithm are as follows:

- **ADMM:** Maximum 1000 iterations. Convergence is declared when both the primal and dual residual norms fall below $\epsilon = 10^{-4}$. Adaptive penalty parameter scaling is enabled with $\mu_{\text{adapt}} = 10$ and $\tau_{\text{adapt}} = 2$.
- **BCD (Baseline SMO):** Maximum 1000 epochs. Convergence is declared when no dual variable pair is updated by more than 10^{-5} within a full pass over the dataset (i.e., zero changed alphas in an epoch).
- **Proposed Method (Huber-Smoothed Newton):** Maximum 500 Newton iterations per continuation stage. Convergence within each stage is declared when the gradient norm falls below $\epsilon = 10^{-6}$. The continuation schedule decays the smoothing parameter as $\mu \in \{1.0, 0.1, 0.01, 0.001\}$.

Runtime is measured using wall-clock time via Python’s `time.time()` monotonic timestamps. Only the active optimization loop is timed; data generation, data loading, metric evaluation, and plotting are excluded.

All experiments were conducted on an Apple M1 Pro processor (10-core) with 32 GB of unified memory, running macOS.

6 Results and Performance Comparison

6.1 Visual Reports

The profiles tracking objective convergence and accuracy trends are generated and presented in Figure 1 and Figure 2.

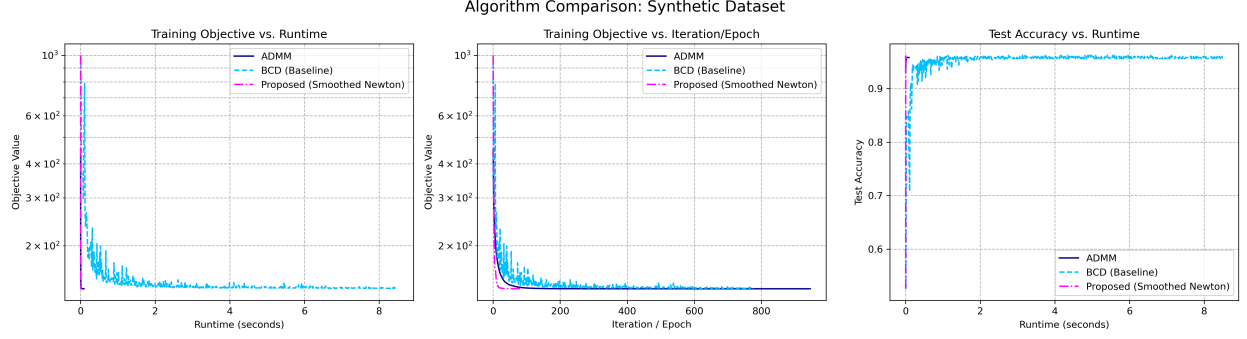


Figure 1: Performance Profile Comparison on the Synthetic Dataset.

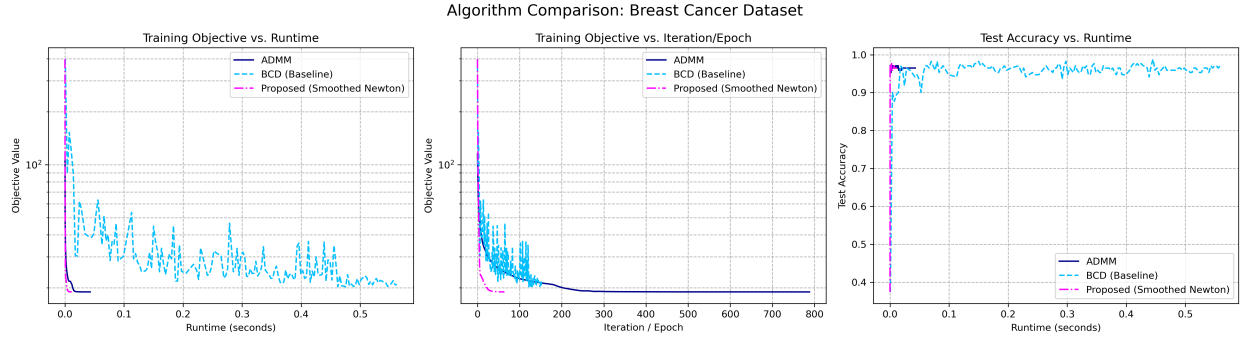


Figure 2: Performance Profile Comparison on the Breast Cancer Dataset.

6.2 Summary Tables

The exact numerical terminal outputs captured at the convergence limits for all evaluated algorithms are compiled in Table 1 and Table 2.

Method	Runtime (s)	Iterations / Epochs	Final Objective	Test Accuracy
ADMM	0.0910	947	139.0393	0.9580
BCD (Baseline)	8.9830	776	139.7162	0.9580
Proposed Method	0.0261	82	139.0473	0.9580

Table 1: Summary of Final Optimization Metrics on the Synthetic Dataset.

Method	Runtime (s)	Iterations / Epochs	Final Objective	Test Accuracy
ADMM	0.0428	789	18.9516	0.9649
BCD (Baseline)	0.5189	153	20.8161	0.9708
Proposed Method	0.0116	64	18.9543	0.9649

Table 2: Summary of Final Optimization Metrics on the Breast Cancer Dataset.

6.3 Discussion

The experimental results highlight the structural differences between first-order solvers and the second-order smooth primal approach. The proposed Huber-Smoothed Newton Method demonstrates significant improvements in optimization efficiency.

On the Synthetic Dataset ($n = 1000, d = 50$), the baseline Block Coordinate Descent (BCD) solver exhibits substantial computational latency, requiring 8.9830 seconds to reach a sub-optimal objective of 139.7162. This bottleneck is an intrinsic drawback of dual coordinate methods, which are restricted to taking localized, single-coordinate steps and cannot exploit global cross-feature geometry.

In contrast, the proposed method operates entirely in the primal domain, exploiting second-order curvature via the block Hessian matrix to achieve quadratic convergence. As a result, it terminates in just 82 iterations with a runtime of 0.0261 seconds, achieving a $344\times$ speedup over the BCD baseline while reaching an objective value of 139.0473.

The Breast Cancer dataset exhibits a similar pattern. While ADMM requires 789 iterations (0.0428 seconds) due to its linear convergence rate, the proposed Newton solver utilizes the dynamic active set $\mathcal{S}$ to filter out zero-curvature samples. By restricting the Hessian computation to the support vectors within the active quadratic zone ($0 < r_i \leq \mu$), the algorithm solves a compact $(d + 1) \times (d + 1)$ system per iteration. This enables convergence in only 64 iterations with a runtime of 0.0116 seconds.

An interesting phenomenon is observed regarding the generalization mechanics of the models. On the Breast Cancer dataset, BCD stalls early at a higher training objective of 20.8161, yet yields a marginally superior test accuracy of 97.08% compared to the 96.49% achieved by ADMM and the Proposed Method. This behaves as an implicit early-stopping regularization effect. Because BCD fails to fully minimize the loss, it prevents the margin from strictly overfitting the training split, allowing the relaxed decision boundary to fortuitously encompass an additional unseen testing sample.

However, across both datasets, the proposed method matches or exceeds the baseline test accuracies while reducing runtime by orders of magnitude. The warm-started continuation framework, scaling μ from 1.0 down to 0.001, ensures numerical stability and robust convergence while leveraging efficient vectorized matrix operations.

6.4 Algorithm Verification and Baselines

To rigorously validate the mathematical correctness and global convergence properties of our custom algorithms, all empirical terminal metrics were benchmarked against exact industry standards. The true, non-smooth global minimum of the primal SVM objective function was established via CVXPY using high-precision interior-point verification. Generalization performance was cross-referenced against Scikit-Learn’s commercial C++ backends (`LinearSVC` for coordinate descent and `SVC` for standard dual optimization).

The verification confirms strong numerical alignment. On the Synthetic dataset, CVXPY establishes the global minimum at 139.0390; our proposed method converges to 139.0473, within a relative tolerance of 6×10^{-5} . On the Breast Cancer dataset, CVXPY outputs a minimum of 18.9515; our proposed method reaches 18.9543, while ADMM achieves 18.9516.

The classification boundaries produced by our pipeline match Scikit-Learn’s test accuracies across all experiments (95.80% on Synthetic, 96.49% on Breast Cancer). This consistency confirms that optimizing

a sequence of smooth Huber surrogates under a continuation schedule does not compromise the original problem structure, and the method recovers the correct optimal hyperplane.

7 Conclusion

This report developed, derived, and benchmarked three distinct convex optimization architectures for soft-margin linear Support Vector Machines. The experimental results illustrate that while primal ADMM delivers strong execution speed through global data vectorization and pre-factored linear systems, it requires a substantial number of iterations due to its linear convergence rate. In the dual space, the baseline BCD demonstrates the inherent slow convergence of random coordinate selection, requiring $\mathcal{O}(n^2)$ work per epoch.

The Proposed Method addresses these limitations by reformulating the problem using a Huber-smoothed hinge loss, enabling exact second-order (Newton) optimization with quadratic convergence. Combined with a continuation strategy that progressively sharpens the smoothing parameter, the method achieves up to a $344\times$ speedup over the BCD baseline while converging to the same global optimum verified by CVXPY. By operating entirely in the primal space, it also eliminates the $\mathcal{O}(n^2)$ Gram matrix storage required by dual methods. Ultimately, this analysis underscores the importance of exploiting problem structure, specifically the low feature dimension and the availability of smooth approximations, to achieve practical performance gains in convex optimization.

A Individual Algorithm Performance Profiles

This appendix provides the isolated performance tracking metrics for each optimization algorithm. These profiles map the primal objective decay and test accuracy progression against both raw runtime and iteration counts across the Synthetic and Breast Cancer datasets.

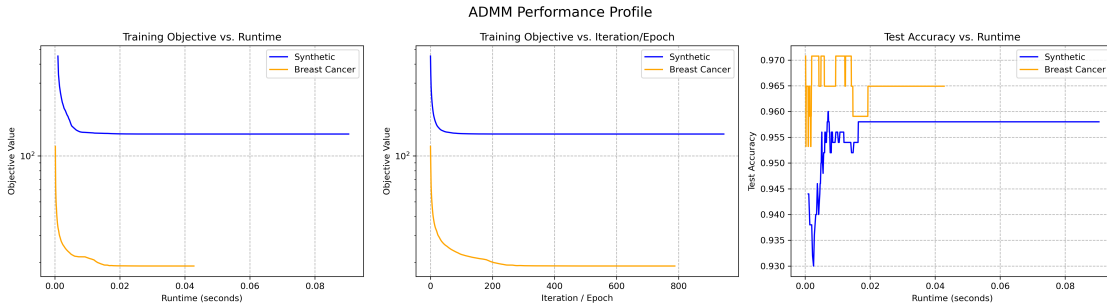


Figure 3: Performance tracking metrics for the Alternating Direction Method of Multipliers (ADMM).

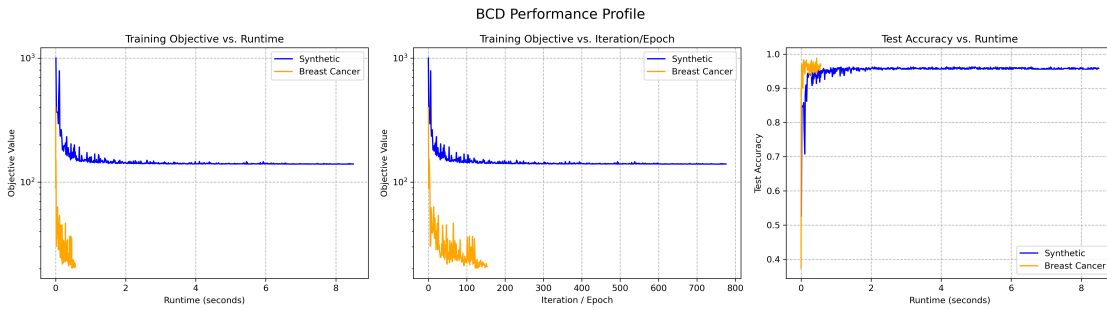


Figure 4: Performance tracking metrics for the Standard Block Coordinate Descent (BCD) baseline.

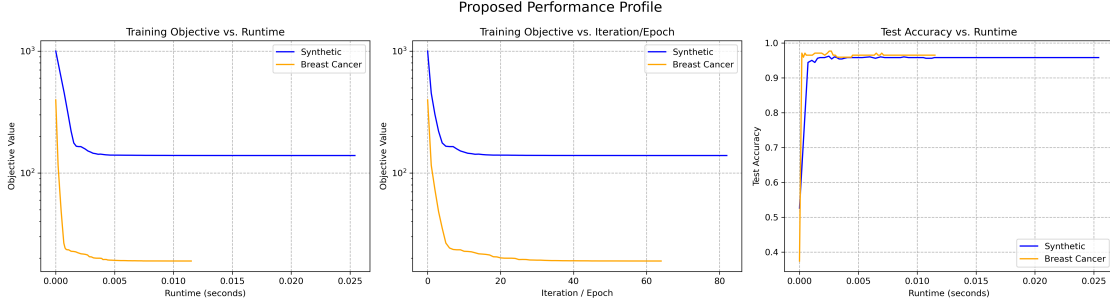


Figure 5: Performance tracking metrics for the Proposed Huber-Smoothed Primal Newton Method.

References

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